Chayank Bhargava

Chennai, India | +91 98107 86558 | chayank.bhargava2023@vitstudent.ac.in

EDUCATION

Vellore Institute of Technology **Chennai, India**
B.Tech., Computer Science *8/2023 -7/2027*
CGPA: 8.83

WORK EXPERIENCE

Annadaata: Autonomous Rover & Drone System for Smart Agriculture, Smart India Hackathon (SIH) Finalist *12/2025*

- Led automation design for autonomous agricultural rover, coordinating control across multiple electromechanical subsystems.
- Designed and integrated a solar tracking "sunflower" mechanism to optimise energy harvesting and extend autonomous operation time.
- Developed an automated sprinkler subsystem enabling controlled water dispersion based on predefined field coverage logic.
- Contributed to rover's electrical architecture, power distribution, motor interfacing, actuator control, safety considerations.
- Supported system-level integration to ensure reliable coordination between mechanical, electrical, and control components.

Cricsolve (Personal Project) *11/2025 – Present*

- Developed a full-stack daily web-based puzzle game using React, TypeScript, and Tailwind CSS, featuring a custom attribute-based game engine.
- Implemented search and filtering logic to compare player attributes (country, role, ICC ranking) against a daily mystery target with real-time UI feedback.
- Designed and maintained scalable JSON-based database of 1,000+ international cricketers across Test, ODI, and T20 formats.
- Implemented state management and persistence using React Hooks and LocalStorage to track game state, progress, and win streaks across sessions.

VIT Chennai: Code for a Cause Competition *01/2024* **Chennai, India**

- Secured 3rd rank in the coding competition focused on solving real-world problems using programming skills.
- Developed innovative solutions and algorithms to address various challenges posed during the competition using Python and Java, including optimization problems, data analysis tasks, and system design issues.
- Collaborated with a team to enhance problem-solving and coding efficiency, participating in pair programming exercises.
- Presented the final project to a panel of judges, detailing the approach, implementation, and impact of the solution.

Knotters (Start-up) *06/2024* **Delhi, India**

- Completed an internship with a dynamic start-up focused on innovative tech solutions in the e-commerce and logistics sector.
- Assisted in the development of web applications using HTML, CSS, and JavaScript, contributing to both the frontend and backend components of the projects.
- Contributed to the backend development using Python, developing RESTful APIs and integrating third-party services such as payment gateways and logistics providers.
- Participated in daily meetings, sprint planning sessions, and code reviews, receiving mentorship from senior developers.
- Documented code for the projects, ensuring maintainability and ease of understanding for future developers.

Programming Languages Proficiency *09/2023-Present* **Remote**

- Highly proficient in four computer programming languages: Python, Java, C, and C++.
- Completed a course on Python from DevTown, enhancing scripting, automation, data analysis, and web development skills.
- Applied programming knowledge in various projects and competitive coding platforms, developing applications such as web crawlers, data visualizations, and game development projects.
- Participated in coding competitions and hackathons, consistently achieving high rankings and receiving recognition for innovative solutions and efficient coding practices.
- Contributed to open-source projects and collaborated with other developers on platforms such as GitHub, enhancing teamwork and community engagement skills.

SKILLS

- Problem Solving, Collaboration
- Communication, Project Management
- UI/UX Design
- Computer Networks, Operating Systems, Computer Architecture, Compiler Design
- **Programming Languages**: Python, Java, C, C++
- **Web Development**: HTML, CSS, JavaScript
- **Frameworks/Libraries**: Node.js, React,
- **Backend Development**: Django, Flask, Ruby on Rails
- **Database Technologies**: SQL, PostgreSQL, MongoDB
- **API Development**: RESTful APIs,